

### SCQ (Single Correct Type) :

- If  $\alpha, \beta$  are eccentric angles of the extremities of a focal chord of an ellipse, then eccentricity of the ellipse is

(A)  $\frac{\cos \alpha + \cos \beta}{\cos(\alpha + \beta)}$  (B)  $\frac{\sin \alpha - \sin \beta}{\sin(\alpha - \beta)}$  (C)  $\sec \alpha + \sec \beta$  (D)  $\frac{\sin \alpha + \sin \beta}{\sin(\alpha + \beta)}$
- A series of concentric ellipses  $E_1, E_2, \dots, E_n$  are drawn such that  $E_n$  touches the extremities of the major axis of  $E_{n-1}$  and the foci of  $E_n$  coincide with the extremities of minor axis of  $E_{n-1}$ . If the eccentricity of the ellipses is independent of  $n$ , then the value of the eccentricity, is

(A)  $\frac{\sqrt{5}}{3}$  (B)  $\frac{\sqrt{5}-1}{2}$  (C)  $\frac{\sqrt{5}+1}{2}$  (D)  $\frac{1}{\sqrt{5}}$
- The transverse axis of a hyperbola is of length  $2a$  and a vertex divides the segment of the axis between the centre and the corresponding focus in the ratio  $2 : 1$ , the equation of the hyperbola is :

(A)  $4x^2 - 5y^2 = 4a^2$  (B)  $4x^2 - 5y^2 = 5a^2$  (C)  $5x^2 - 4y^2 = 4a^2$  (D)  $5x^2 - 4y^2 = 5a^2$
- If AB is a double ordinate of the hyperbola  $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$  such that  $\triangle OAB$  (O is the origin) is an equilateral triangle, then the eccentricity 'e' of the hyperbola satisfies

(A)  $e > \sqrt{3}$  (B)  $1 < e < 2\frac{2}{\sqrt{3}}$  (C)  $e = \frac{2}{\sqrt{3}}$  (D)  $e > \frac{2}{\sqrt{3}}$
- An ellipse whose major axis is parallel to X –axis such that the segments of the focal chords are 1 and 3 units. The lines  $ax + by + c = 0$  are the chords of the ellipse such that  $a, b, c$  are in A.P. and bisected by the point at which they intersect. The equation of its auxiliary circle is  $x^2 + y^2 + 2\alpha x + 2\beta y - 2\alpha - 1 = 0$  then \_\_\_\_\_.  
 Equation of the auxiliary circle is

(A)  $x^2 + y^2 - 2x + 4y + 1 = 0$  (B)  $x^2 + y^2 + 2x + 2y - 3 = 0$   
 (C)  $x^2 + y^2 + 2x + 4y + 1 = 0$  (D)  $x^2 + y^2 - 4x + 2y - 3 = 0$

### MCQ (One or more than one correct) :

- A point moves such that the sum of the squares of its distances from the two sides of length 'a' of a rectangle is twice the sum of the squares of its distances from the other two sides of length 'b'. The locus of the point can be :

(A) a circle (B) an ellipse (C) a hyperbola (D) a pair of lines

7. Which of the following equations in parametric form can represent a hyperbolic profile, where 't' is a parameter.
- (A)  $x = \frac{a}{2} \left( t + \frac{1}{t} \right)$  &  $y = \frac{b}{2} \left( t - \frac{1}{t} \right)$  (B)  $\frac{tx}{a} - \frac{y}{b} + t = 0$  &  $\frac{x}{a} + \frac{ty}{b} - 1 = 0$
- (C)  $x = e^t + e^{-t}$  &  $y = e^t - e^{-t}$  (D)  $x^2 - 6 = 2 \cos t$  &  $y^2 + 2 = 4 \cos^2 \frac{t}{2}$
8. Let  $C_1 : 9x^2 - 16y^2 - 18x + 32y - 23 = 0$  and  $C_2 : 25x^2 + 9y^2 - 50x - 18y + 33 = 0$  are two conics then
- (A) eccentricity of  $C_1$  is  $\frac{5}{4}$ . (B) eccentricity of  $C_2$  is  $\frac{5}{3}$ .
- (C) area of the quadrilateral with vertices at the foci of the conics is  $\frac{8}{9}$ .
- (D) latus rectum of  $C_1$  is greater than latus rectum of  $C_2$ .
9. A hyperbola centred at C has one focus at P(6, 8). If its directrices are  $3x + 4y + 10 = 0$  and  $3x + 4y - 10 = 0$ , then \_\_\_\_.
- (A) CP = 10 (B) eccentricity =  $\sqrt{5}$
- (C) CP = 8 (D) eccentricity =  $\frac{\sqrt{5}}{2}$
10. Two ellipses  $\frac{x^2}{\cos^2 \alpha} + \frac{y^2}{\sin^2 \alpha} = 1$  and  $\frac{x^2}{\sin^2 \alpha} + \frac{y^2}{\cos^2 \alpha} = 1 \left( 0 < \alpha < \frac{\pi}{4} \right)$  intersect at four points P, Q, R, S then which of the following statement(s) is /are true?
- (A) PQRS is a square with length of the side  $\sin 2\alpha$
- (B) PQRS lie on a circle whose centre is origin and with radius  $\frac{\sin 2\alpha}{\sqrt{2}}$
- (C) eccentricity of the two given ellipses are same
- (D) there are two points on  $\frac{x^2}{\sin^2 \alpha} + \frac{y^2}{\cos^2 \alpha} = 1$  whose reflection in  $y = x$  lie on the same ellipse

### Comprehension Type Question:

#### Comprehension # 1

Paragraph for question nos. 11 to 13

The graph of the conic  $x^2 - (y - 1)^2 = 1$  has one tangent line with positive slope that passes through the origin. the point of tangency being (a, b). Then

11. The value of  $\sin^{-1} \left( \frac{a}{b} \right)$  is

- (A)  $\frac{5\pi}{12}$  (B)  $\frac{\pi}{6}$  (C)  $\frac{\pi}{3}$  (D)  $\frac{\pi}{4}$

12. Length of the latus rectum of the conic is  
 (A) 1 (B)  $\sqrt{2}$  (C) 2 (D) none
13. Eccentricity of the conic is  
 (A)  $\frac{4}{3}$  (B)  $\sqrt{3}$  (C) 2 (D) none

**Numerical based Questions :**

14. Common tangents are drawn to the parabola  $y^2 = 4x$  & the ellipse  $3x^2 + 8y^2 = 48$  touching the parabola at A & B and the ellipse at C & D, then the area of the quadrilateral ABCD is  $\lambda\sqrt{2}$  the  $\lambda$  is equal to
15. If two points P & Q on the hyperbola  $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$  whose centre is C be such that CP is perpendicular to CQ &  $a < b$ , then  $\frac{1}{CP^2} + \frac{1}{CQ^2} = \lambda \left( \frac{1}{a^2} - \frac{1}{b^2} \right)$  where  $\lambda$  is :
16. If the distance between the centres of the hyperbolas :  
 $x^2 - 16xy - 11y^2 - 12x + 6y + 21 = 0$  ..... (i)  
 $9x^2 - 16y^2 - 18x - 32y - 151 = 0$  ..... (ii)  
 is d then  $125d^2 = \dots\dots\dots$
17. If  $(a^2, a - 2)$  be a point interior to the region of the parabola  $y^2 = 2x$  bounded by the chord joining the points (2, 2) and (8, -4), then the number of all possible integral values of a is :
18. If a tangent of slope 2 of the ellipse  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$  is normal to the circle  $x^2 + y^2 + 4x + 1 = 0$ , then maximum value of ab is \_\_\_\_\_.
19. If the length of the latus rectum of a standard hyperbola of eccentricity 2 is equal to the limit of the series  $\frac{3}{1^2} + \frac{5}{1^2 + 2^2} + \frac{7}{1^2 + 2^2 + 3^2} + \dots$  and r is the radius of the director circle of its conjugate hyperbola, then  $r^2 = \dots\dots\dots$ .
20. Let  $C_1 : x^2 - y^2 = 5$  and  $C_2 : x^2 + y^2 - 8y + 3 = 0$  be the equations of a hyperbola and a circle respectively. The curves  $C_1$  and  $C_2$  touch each other at  $(\pm 3, 2)$ .  $P(0, 4 - \sqrt{13})$  is a point on the curve  $C_2$ . Let a line through P meet  $C_1$  at m number of points and  $C_2$  at n number of points. If  $(m + n) = 3$ , then the number of such straight lines is \_\_\_\_\_.
21. Coordinates of the vertices B and C are (2, 0) and (8, 0) respectively. The vertex A is varying in such a way that  $4 \tan \frac{B}{2} \tan \frac{C}{2} = 1$ . If the locus of A is an ellipse then the length of its semi major axis is \_\_\_\_\_.

**Matrix Match Type :**

22. Match the following:

For the ellipse  $3x^2 - 6x + 3y^2 - 9y = \frac{4}{169} (5x + 12y + 6)^2$

| Column-I                         | Column-II          |
|----------------------------------|--------------------|
| (a) Length major axis            | (p) $\frac{24}{5}$ |
| (b) Length minor axis            | (q) $\frac{16}{5}$ |
| (c) Length of Latus Rectum       | (r) $\frac{16}{3}$ |
| (d) Distance between directrices | (s) $\frac{72}{5}$ |
|                                  | (t) $\frac{48}{5}$ |

(A)  $a \rightarrow r$ ;  $b \rightarrow q$ ;  $c \rightarrow r$ ;  $d \rightarrow s$

(B)  $a \rightarrow t$ ;  $b \rightarrow q$ ;  $c \rightarrow r$ ;  $d \rightarrow s$

(C)  $a \rightarrow t$ ;  $b \rightarrow s$ ;  $c \rightarrow r$ ;  $d \rightarrow s$

(D)  $a \rightarrow t$ ;  $b \rightarrow q$ ;  $c \rightarrow q$ ;  $d \rightarrow s$

**Subjective Type Questions :**

23. If S and H be the foci of an ellipse and any point A be taken on the curve and the chords ASB, BHC, CSD and DHE be drawn and eccentric angles of A, B, C, D, E, ..... be  $\theta_1, \theta_2, \theta_3, \theta_4, \dots$ . Prove that  $\tan \frac{\theta_1}{2} \tan \frac{\theta_2}{2} = \cot \frac{\theta_2}{2} \cot \frac{\theta_3}{2} = \tan \frac{\theta_3}{2} \tan \frac{\theta_4}{2} = \dots$
24. A variable point P on an ellipse of eccentricity e is joined to its foci S, S'. Prove that the locus of the incentre of  $\triangle PSS'$  is an ellipse of eccentricity  $\sqrt{\frac{2e}{1+e}}$ .
25. Let P and Q be two points on the ellipse  $x^2 + 4y^2 = 16$  whose eccentric angles are  $\frac{\pi}{4}$  and  $\frac{3\pi}{4}$  respectively. The tangent at P and the normal at Q cut each other at R and the normal at Q cuts the ellipse again at M. Find the area of the triangle PRM.
26. A rectangular hyperbola, with centre C, is intersected by a circle of radius r in four points P, Q, R and S. Prove that  $CP^2 + CQ^2 + CR^2 + CS^2 = 4r^2$